

# Assignment Testing

CSC354 · Machine Learning

|                           |                           |                             |                         |                        |
|---------------------------|---------------------------|-----------------------------|-------------------------|------------------------|
| TOTAL MARKS<br><b>100</b> | DURATION<br><b>30 min</b> | DIFFICULTY<br><b>MEDIUM</b> | THRESHOLD<br><b>60%</b> | QUESTIONS<br><b>10</b> |
|---------------------------|---------------------------|-----------------------------|-------------------------|------------------------|

|              |            |         |      |
|--------------|------------|---------|------|
| STUDENT NAME | STUDENT ID | SECTION | DATE |
|--------------|------------|---------|------|

## INSTRUCTIONS

- Read all questions carefully before answering.
- All questions are compulsory unless stated otherwise.
- Total time allowed: 30 minutes.
- No electronic devices allowed during the quiz.
- Write clearly and legibly in the space provided.

## Questions

### Assessment Questions

- 1 In the context of machine learning, which algorithm is best suited for environments with complex dynamics and requires a system to learn optimal strategies through trial and error?

[10 marks | Applying | medium]

- ☐ A Option A: Linear Regression
- ☐ B Option B: Deep Reinforcement Learning
- ☐ C Option C: K-Nearest Neighbors (KNN)
- ☐ D Option D: Principal Component Analysis (PCA)

Student Answer: &

- 2 Which algorithm is commonly used to train agents in environments with complex dynamics for reinforcement learning?

[10 marks | Applying | medium]

- ☐ A Supervised Learning
- ☐ B Reinforcement Learning using Q-learning
- ☐ C Regression Analysis
- ☐ D Genetic Algorithms

Student Answer: &

- 3 Which of the following algorithms is commonly used in deep learning for supervised tasks?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Convolutional Neural Networks (CNN)
- ☐ C K-Means Clustering
- ☐ D Principal Component Analysis (PCA)

Student Answer: &

4 Which algorithm is most suitable for learning policies in environments with complex dynamics?

[10 marks | Applying | medium]

- ☐ A Linear regression
- ☐ B Decision trees
- ☐ C Deep Q-Network (DQN)
- ☐ D Naïve Bayes classifier

Student Answer: &

5 In the context of machine learning, which algorithm is best suited for environments with complex dynamics and requires a model to learn optimal policies through trial-and-error interactions?

[10 marks | Applying | medium]

- ☐ A Option A: Linear Regression
- ☐ B Option B: Deep Reinforcement Learning
- ☐ C Option C: K-Means Clustering
- ☐ D Option D: Principal Component Analysis (PCA)

Student Answer: &

6 In the context of machine learning, which algorithm is commonly used to solve problems in environments with complex dynamics by interacting and learning from its environment?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Decision Trees
- ☐ C Reinforcement Learning
- ☐ D Naïve Bayes Classifier

Student Answer: &

7 Which algorithm is often used for learning policies in complex environments where the agent must learn from rewards and penalties?

[10 marks | Applying | medium]

- ☐ A Decision Trees
- ☐ B Neural Networks with Reinforcement Learning
- ☐ C Linear Regression Models
- ☐ D Support Vector Machines

Student Answer: &

8 Which algorithm is most suitable for solving problems where the environment's dynamics are complex and change over time?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Decision Trees
- ☐ C Reinforcement Learning
- ☐ D Naïve Bayes Classifier

Student Answer: &

9

Which algorithm is best suited for environments with complex dynamics and requires the model to learn optimal policies through trial and error?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Decision Trees
- ☐ C Q-Learning (a type of Reinforcement Learning)
- ☐ D Naive Bayes Classifier

Student Answer: &

10

Which algorithm is commonly used for training agents to perform tasks in complex, dynamic environments?

[10 marks | Applying | medium]

- ☐ A Linear Regression
- ☐ B Decision Trees
- ☐ C Q-Learning (Reinforcement Learning)
- ☐ D Naive Bayes Classifier

Student Answer: &